SOFTWARE REQUIREMENTS SPECIFICATION

MindBridge

Multimodal Intelligence Learning Platform

A web application that presents middle-school curriculum concepts through configurable learning pathways and evaluates learning outcomes.

Document Details

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Document Control

Version	Date	Author	Description
0.1	07-Oct-2026	Project team	Initial draft
1.0	07-Oct-2026	Project team	Full SRS draft
1.1	10-Oct-2026	Project team	Revised scope and requirements; clarified scoring, baseline testing, adaptation, roles, privacy, and validation

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1.Introduction:

1.1 Purpose

This document specifies the functional, non-functional, and technical requirements for MindBridge. It is intended for the four-person project team, Pinnacle mentors, and participating school representatives. For the Mini Project course of the V semester, only two third students work on this project.

1.2 Problem Statement

Students may engage with curriculum concepts through different representations, such as language, reasoning, visualizations, sound, interactive activities, and real-world observation. The application will let students explore a learning preference profile and receive curriculum content through a recommended pathway. It will measure learning performance separately from preference and use that evidence to guide content recommendations.

1.3 Scope

In Scope	Out Of Scope
Middle-school students, primarily Classes 6–8	Primary and high-school grades
Eight pathways: Linguistic, Logical-Mathematical, Spatial/Visual, Musical, Bodily-Kinesthetic, Naturalistic, Interpersonal and Intrapersonal mentioned in Gardner Theory	No other intelligence types mentioned in other theories
Responsive Web-based interactive learning platform	Native mobile application
Tamil Nadu State Board curriculum for classes 6-8	No other boards/ curriculum like CBSE, ICSE or other State Boards
Student, teacher, administrator roles; parent/guardian access	Payments, subscriptions, live video classes
Preference profile, intelligence-based content, concept quizzes, progress reports	Automated AI publishing without teacher review

1.4 Definitions

Term	Meaning
Learning preference profile	Relative scores across the eight supported learning pathways; not a diagnosis.
Primary Intelligence	Intelligence with the highest normalized preference score.
Secondary Intelligence	Intelligence with the second-highest normalized preference score.
Concept	A defined curriculum learning unit, e.g., Photosynthesis.
Resource	Videos, pictures, equations, activities used to teach students a particular concept.
Pre-assessment	Concept-level assessment taken before the learning resource.
Learning gain	Normalized change between comparable concept pre-test and post-test scores.
Engagement indicators	Completion, time on task, retries, interaction and optional student feedback.
Mastery	Meeting the configured post-assessment criterion for a concept.

1.5 Conceptual framing

The system must not present its output as a clinical, psychological, or scientifically validated intelligence diagnosis. The system uses terms such as learning preference, learning pathway, engagement, and profile. A preference is not treated as proof that a student learns best through that intelligence type.

1.6 Background and references

The team has made background research on Howard Gardner's Multiple Intelligences, review of Tamil Nadu textbooks, and visited one Government Middle School and one private school. Online questionnaires may inform original question design but it should not be considered as validated instruments.

2. Overall Description

2.1 Product perspective

The product is a responsive web application with connected assessment, learning, and evaluation modules. Teacher-managed curriculum resources are stored separately from application code.

2.2 Product workflow

```
Student registration/login
        ↓
Learning preference assessment
        ↓
Eight-pathway profile (primary + secondary)
        ↓
Teacher-approved pathway-based content
        ↓
Student learns the curriculum concept in the primary intelligence type
        ↓
Concept baseline quiz
        ↓
Concept post-quiz + engagement feedback + class test
        ↓
Learning gain / mastery analysis
        ↓
Continue primary pathway OR consider blended pathway (primary + secondary)
```

2.3 User classes and permissions

Role	Permitted Actions
Student	Complete assessment, view own profile, learn, take quizzes, view own progress and provide feedback.
Teacher	View assigned classes/students, upload or draft resources, review/approve resources if granted permission, record class-test marks, view reports and provide observations.
Administrator	Manage accounts, roles, schools/classes, subjects, chapters, concepts, assessment configuration and system settings.
Parent/guardian (deferred unless approved)	If enabled, view only linked child's permitted progress and submit optional observations.
Content creator (permission, not necessarily separate role)	Create and tag draft resources; publishing requires an authorized approver.

2.4 Operating environment

- Modern versions of Chrome, Edge, Brave or Firefox.
- Responsive use on computers and tablets; mobile-browser support is desirable.
- Internet connection required for the MVP; optimize media for limited bandwidth.
- Hosting will be a school authorized cloud service, with HTTPS for deployment.

2.5 Constraints and assumptions

- Content must align with the selected school curriculum and learning objectives.
- Students are minors; collect only necessary information and use appropriate school/parent consent for concepts.
- Teachers will provide or approve content and may enter classroom test results.
- The initial release will prioritize a small, complete workflow over broad content coverage.
- Tamil/English support should be implemented

3 Specific Requirements

3.1 Supported Intelligence Types

Intelligence Type	Respective Formats
Linguistic	Stories, explanations, reading passages, vocabulary and analogies.
Logical-Mathematical	Patterns, puzzles, reasoning, cause-effect and structured problems.
Spatial/Visual	Diagrams, images, maps, animations, graphs and visualizations.
Musical	Rhymes, songs, audio explanations and rhythm-based memory aids.
Bodily-Kinesthetic	Interactive activities, simulations, drag-and-drop and demonstrations.
Naturalistic	Observation, classification, environmental examples and nature-based activities.
Interpersonal	Collaborative learning, group discussions, peer teaching and role-play.
Intrapersonal	Self-reflection, individual goal-setting, personal journaling and independent reflection

3.2 Module 1 — Pre-Assessment

3.2.1 Description

After first registration, the student is directed to a learning preference assessment. The module generates normalized scores for all eight supported pathways, displays the profile, and records primary and secondary pathways. The pre-assessment can be retaken under defined rules.

3.2.2 Question types

3.2.2.1 Preference/self-report questions: students choose formats they would prefer; there is no right or wrong answer.

3.2.2.2 Scenario questions: students select how they would approach a learning situation; each option maps to one or more intelligence types.

3.2.2.3 Teacher/parent observations authorized teacher records observations, including parent-reported ones, and labels the source. Parents do not log in.

3.2.3 Scoring rules

Preference scores shall be calculated primarily from selected options and their intelligence mappings. Response time is recorded as an additional factor during exact tie of intelligence profile and does not directly penalize a student for answering slowly, i.e., response time never changes preference scores.

Normal scoring: Calculate and normalize each intelligence score using only option contributions.

Tie-breaking: If the top two scores are tied use response time as a secondary factor to help determine the primary and secondary intelligences.

```
For each intelligence k:

RawPreference(k) = Σ [option contribution(k)]

NormalizedPreference(k) = 100 × RawPreference(k) / MaximumPossiblePreference(k)

Primary Intelligence = highest normalized preference scores
Secondary Intelligence = second-highest normalized preference
score
```

The implementation must define how unanswered items, multiple mappings, repeated attempts, and zero denominators are handled. If the top two intelligence scores are tied or fall within a configurable near-tie margin, response time shall be used as a secondary factor to help determine the primary and secondary intelligences. If response time does not resolve the tie, the system shall use a predefined, deterministic intelligence ordering as the final tie-breaker. Response time shall not affect normal scoring or penalize students who take longer to answer.

3.2.4 Functional requirements

ID	Requirement	Priority
PRE-F1	The system shall support student self-registration and admin provided teacher accounts.	High
PRE-F2	A newly registered student shall be directed to the pre-assessment before personalized learning.	High
PRE-F3	Each question option shall map to one or more of the eight intelligence types.	High
PRE-F4	The assessment shall support self-report, scenario-based questions.	High
PRE-F5	The system shall save each response and response time.	High
PRE-F6	The backend shall aggregate intelligence preference contributions using configured mappings.	High
PRE-F7	The system shall normalize all eight intelligence scores to a 0–100 scale, handling missing responses safely.	High
PRE-F8	The system shall rank eight intelligences and identify primary and secondary intelligences using the documented score rule mentioned in section 3.2.3	High
PRE-F9	The system shall display eight scores using an accessible chart and text summary.	Medium
PRE-F10	The system may collect optional teacher/parent observations. Both teacher parent observations are entered by the teacher.	Low
PRE-F11	The system shall store profile history and make the current profile available to content recommendation.	High
PRE-F12	Authorized administrators shall manage questions, and mappings	High
PRE-F13	The system shall allow reassessment under defined rules without silently overwriting historical profiles.	Medium

3.2.5 Non-functional and technical requirements

3.2.5.1 Questions and instructions must use age-appropriate language.

3.2.5.2 The next question should load within two seconds under normal test conditions.

3.2.5.3 Responses should be saved incrementally and recover safely after a connection interruption.

3.2.5.4 The assessment should target a duration of 20–30 minutes, subject to usability testing.

3.2.5.5 The timer must not reset on refresh; server timestamps should be used where appropriate.

3.2.5.6 Question coverage across all eight intelligences must be sufficient.

3.2.5.7 Student responses and profiles must be accessible only to authorized users.

3.3 Module 2 — Specialized Content Learning

3.3.1 Description

The module presents a curriculum concept using resources associated with the student's primary intelligence type. The learning objective and core curriculum content must remain consistent across intelligence types; representation and activity may vary.

3.3.2 Curriculum hierarchy

```
School → Class/Grade → Subject → Chapter/Unit → Concept → Learning resources
```

3.3.3 Example: Photosynthesis

Intelligence Type	Example Resource
Linguistic	Simple explanation or story with key terms.
Logical-Mathematical	Flowchart and cause-effect reasoning about inputs and outputs.
Spatial/Visual	Animated diagram of sunlight, water and carbon dioxide producing glucose and oxygen.
Musical	Teacher-reviewed rhyme, song or audio mnemonic.
Bodily-Kinesthetic	Drag-and-drop simulation arranging process inputs and outputs.
Naturalistic	Guided observation of plants and discussion of growth conditions.
Interpersonal	Students work in group to explain each part of the photosynthesis concept
Intrapersonal	Student learning the photosynthesis concept independently and reflect it themselves.

3.3.4 Concept and resource requirements

3.3.4.1 Each concept shall include a learning objective, curriculum reference, difficulty level, and one or more resources.

3.3.4.2 Each resource shall be tagged with grade, subject, chapter, concept, intelligence type, format, language, version, creator, and approval state.

3.3.4.3 A concept shall have a short quiz linked to its learning objectives.

3.3.4.4 The system shall record concept status: not started, in progress, completed, or needs review.

3.3.4.5 Students may resume a concept without losing progress.

3.3.4.6 The content-selection service shall use the current primary intelligence type by default and support a secondary/blended intelligence type when recommended by the teacher.

3.3.5 Content approval workflow

```
Draft / Uploaded -> Pending Review -> Approved -> Published -> Rejected / Needs Revision → Draft
```

Only approved and published resources will be shown to students. AI-assisted resources, if added, must be marked as drafts and reviewed by an authorized teacher before publication. The system shall retain approval status, reviewer, timestamp, and resource version.

3.3.6 Design Decision for MVP:

Module 1 - Pre-assessment: Continue calculating the preference profile across all eight intelligence types, as specified in your current SRS.

Module 2 - Content learning MVP: Initially prepare and deliver content for only the three intelligence types (Visual, Linguistic, Logical-Mathematical)

3.3.7 Functional requirements

ID	Requirement	Priority
LRN-F1	The student shall browse their subject, chapter and concept.	High
LRN-F2	The system shall provide approved content for their current primary intelligence type.	High
LRN-F3	The system shall support content for three intelligence type as per the design decision.	High
LRN-F4	Content authoring/management shall be separate from the student learning interface.	High
LRN-F5	Authorized users shall upload, tag, edit, version and archive resources.	High
LRN-F6	The system shall offer secondary/blended intelligence type content when recommended by the teacher.	High
LRN-F7	A concept assessment shall be available after learning content.	High
LRN-F8	Student-visible resources shall require approved/published status.	High
LRN-F9	The system shall support appropriate text, image, audio, video and interactive formats.	Medium
LRN-F10	Students shall be able to mark a concept complete when its completion conditions are met.	Medium
LRN-F11	The system shall save progress and allow the student to resume.	Medium
LRN-F12	The system shall retain the concept resources for students to learn again.	Medium

3.3.8 Non-functional requirements

3.3.8.1 Content must match the selected curriculum and be reviewed for age appropriateness.

3.3.8.2 Media should be optimized for slow school networks; provide text alternatives where feasible.

3.3.8.3 Images require alt text; audio/video should include transcripts or captions where feasible.

3.3.8.4 Adding subjects and resources should not require changes to application code.

3.3.8.5 Content and metadata should be stored separately from application logic.

3.4 Module 3 — Post-Assessment and Adaptation

3.4.1 Assessment sequence

When learning gain measured for a concept, the system shall use a concept-level baseline assessment before the specialized content and a comparable post-assessment after it.

```
Concept → Baseline quiz → Learning content → Post-quiz → Analysis
```

3.4.2 Recorded data

3.4.2.1 Baseline and post-test scores, quiz version, question responses, attempts and time taken.

3.4.2.2 Concept completion and pathway/resource version used.

3.4.2.3 Student feedback and other engagement indicators.

3.4.2.4 Teacher-recorded classroom assessment marks, with date and subject/concept when known.

3.4.2.5 Profile version used at the time of learning, so later profile changes do not rewrite historical evidence.

3.4.3 Learning gain and mastery

For comparable pre-test and post-test scores on a 0–100 scale, normalized learning gain may be calculated as:

```
Learning Gain = (Post-test score − Pre-test score) / (100 − Pre-test score)
```

Example: pre-test 40, post-test 70 gives (70−40)/(100−40) = 0.50, or 50% normalized gain. If the pre-test score is 100, the denominator is zero; the system shall mark normalized gain as not applicable and report the raw score change instead. Negative gain is valid and must not be silently clamped to zero.

Mastery is a separate decision based on a configurable concept pass mark and confirmed by the teacher. Improvement does not automatically mean mastery. Pre/post assessments should cover the same learning objectives and have comparable difficulty.

3.4.4 Adaptation logic

```
Post-assessment result
        ↓
Check completion, learning time, quiz alignment, attempts and difficulty
        ↓
Sufficient evidence of low learning?
    No → Continue/offer support and gather more evidence
    Yes → Recommend primary + secondary blended pathway
        ↓
Re-teach/retry and reassess
        ↓
Record pathway used and outcome
```

The system shall not automatically switch pathways after a single poor score without checking whether the student completed the content and whether the assessment was suitable. Student or teacher override shall be available. The threshold for mastery and any adaptation rule must be configurable and validated during pilot testing.

3.4.5 Functional requirements

ID	Requirement	Priority
PST-F1	The system shall provide a concept quiz after learning.	High
PST-F2	The system shall record quiz score, attempts, responses and time.	High
PST-F3	Authorized teachers shall be able to enter classroom test scores.	High
PST-F4	The system shall calculate concept mastery using a configured criterion.	High
PST-F5	The system shall calculate learning gain only when comparable concept pre/post scores exist.	High
PST-F6	The system shall record engagement indicators separately from academic scores.	Medium
PST-F7	The system shall evaluate adaptation conditions using documented rules and thresholds.	High
PST-F8	Satisfactory learning shall allow the student to continue to the next concept.	High
PST-F9	When evidence supports adaptation, the system shall recommend secondary/blended content and permit reassessment.	High
PST-F10	Students shall view their own progress in accessible charts and text.	Medium
PST-F11	Teachers shall view authorized student and class reports.	Medium
PST-F12	Students may provide optional feedback on content helpfulness/difficulty.	Low
PST-F13	The system shall keep a history of scores, resource versions, and pathway changes.	High

4 System-Wide Requirements

4.1 User interface

- Child-friendly and responsive screens with clear navigation and large controls.
- Separate views for student, teacher, and administrator; parent access is deferred for the MVP.
- Charts must have labels or textual alternatives; do not rely on colour alone.
- Support keyboard navigation, readable contrast, alt text, and captions/transcripts where feasible.

4.2 Security and privacy

- Use role-based authorization on both frontend and backend; hiding a button is not sufficient access control.
- Passwords must be stored using a modern password-hashing algorithm; do not store plaintext or reversibly encrypted passwords.
- Server-side cookie sessions, HttpOnly, Secure and SameSite=Lax in deployment, session expiry and logout invalidation; no bearer JWTs used.
- Students may access only their own records.
- Teachers may access only assigned classes/students; administrators receive only the access needed for their duties.
- Collect the minimum necessary student data and establish school/parent consent for user testing.
- Define backup, restoration, retention, and deletion procedures for student records.
- Do not expose student profiles publicly or use student data for unrelated purposes.

4.3 Reliability and performance

- Save assessment and quiz submissions reliably; prevent duplicate submissions where practical.
- Core pages should load within two seconds under the team's defined test conditions, excluding large media downloads.
- Provide useful error messages and retry/resume behaviour for intermittent connectivity.
- Perform regular database backups and test restoration before a real user.

4.4 Availability and language

The application should be available during planned school use. English is the initial language unless the team confirms bilingual delivery. Tamil support is an explicit scope decision to resolve before content production.

5 Technical Requirements

5.1 Tech stack Recommendations:

Layer	Starting Choice	Notes
Frontend	React.js, Angular.js	Responsive component-based interface.
Backend	Node.js + Express, FastAPI	Suitable for REST APIs and role-based access.
Database	PostgreSQL	Relational data and constraints for students, assessments, learning resources, and results..
Authentication	Secure cookie/session using HTTP only cookies	A practical option for a browser-based school application. Enforce authorization on the server.
Media	Cloud storage for both development and deployment.	metadata, approval status, version, and storage reference in postgres database.
Tools	Git, GitHub, Postman, issue tracker	Suitable for team collaboration, API testing, and tracking tasks.

5.2 Main data entities

Entry	Purpose
users	Login identity, role based access control, account status.
students	Student-specific profile, grade/class link.
teachers	Teacher details and class assignments.
schools, classes, class_memberships	Institution and class structure.
subjects, chapters, concepts	Curriculum hierarchy and learning objectives.
intelligence_types	The eight supported pathways.
assessment_questions, assessment_options, question_pathway_mappings	40 pre-assessment questions: 24 self-report + 16 scenario based.
assessment_attempts, assessment_responses	Assessment session, responses and timestamps.
intelligence_profiles, profile_scores	Versioned eight intelligence profile and ranks.
learning_resources, resource_versions, resource_reviews	Content, pathway tags, approval state and review history.
quizzes, quiz_questions, quiz_attempts, quiz_responses	support subject-knowledge pre-tests and post-tests separate from intelligence mapping pre-assessment.
progress_records, pathway_events	Concept completion and intelligence type/resource used.
class_test_scores, teacher_feedback, student_feedback	Supplementary assessment and feedback.

5.3 Core relationships

School → Classes → Students

Teachers → Class Assignments → Classes

Subjects → Chapters → Concepts → Learning Resources

Concepts → Quizzes → Quiz Attempts → Students

Students → Assessment Attempts → Assessment Responses

Students → Intelligence Profiles → Profile Scores

Students + Concepts → Progress Records

5.4 Backend services

- Authentication and authorization service.
- Pre-assessment service: question delivery, response persistence, profile scoring.
- Curriculum/content service: browse, tag, approve and recommend resources.
- Quiz service: quiz delivery, scoring, attempts and baseline/post-test association.
- Progress/adaptation service: mastery, learning gain, pathway recommendations and history.
- Reporting service: student progress and class-level summaries.

6 Acceptance Criteria and Test Strategy

6.1 MVP acceptance criteria

- A student can register/login and is directed to the preference assessment when no current profile exists.
- The assessment produces eight normalized scores, primary and secondary intelligence types, and a visible profile chart.
- A student receive approved content associated with the current intelligence type.
- The student can complete a concept baseline quiz, learning content, and post-quiz; scores and attempts persist.
- Learning gain is calculated correctly for valid pre/post scores and handles a pre-test score of 100 safely.
- Low-performance triggers diagnostic checks before a blended-pathway recommendation; manual override is recorded.
- A teacher can approve content, view assigned students' progress, and enter classroom test marks.
- Unauthorized users cannot access other students' profiles or restricted teacher/admin functions.

6.2 Test categories

Category	Examples
Functional	Registration, assessment, scoring, content selection, quizzes, progress and approval workflow.
Scoring	Eight intelligence type mappings, normalization, missing answers, ties, near-ties, zero denominator.
Adaptation	Good score, low score, incomplete content, insufficient time, manual override, blended pathway.
Security	Role boundaries, direct API access, unauthorized student-record access, session expiry.
Usability/accessibility	Student comprehension, tablet layout, keyboard access, chart labels, media alternatives.
Reliability	Connection loss, duplicate submission, refresh/resume, backup restoration.

6.3 User evaluation

User testing should assess usability, completion, engagement, and concept-level performance. Results must be reported with sample size and limitations. The project must not claim that a small user scientifically validates Gardner's theory or proves a universal learning style.

Appendix A — References

Curriculum source: Tamil Nadu school textbooks portal — https://www.tntextbooks.in/p/school-books.html

Questionnaire references used for background and original question design:

- https://psychology.com/tests/multiple-intelligences
- https://alis.alberta.ca/careerinsite/know-yourself/multiple-intelligences-quiz/
- https://www.mentalup.co/blog/multiple-intelligence-test
- https://prismatest.com/en/tests/garner

These online resources are references for question design, not automatic evidence that the resulting MindBridge assessment is a validated instrument.

Appendix B — Requirement traceability

Project Objective	Requirement IDs
Register and assess new student	PRE-F1, PRE-F2, PRE-F5
Eight intelligence type preference profile	PRE-F3, PRE-F6, PRE-F7, PRE-F8, PRE-F9
Personalized approved curriculum content	LRN-F1, LRN-F2, LRN-F3, LRN-F11
Teacher content review	LRN-F5, LRN-F11
Concept quizzes and learning gain	PST-F1, PST-F2, PST-F4, PST-F5
Adaptive primary/secondary pathway	LRN-F6, PST-F7, PST-F9, PST-F13
Teacher monitoring and class marks	PST-F3, PST-F11
Privacy and access control	Sections 2.3 and 4.2

Appendix C — Decision Chosen for SRS

ID	Options	Decision Chosen
OI-1	Initial language: English or English + Tamil?	English for MVP.
OI-2	Content authoring for MVP	Manually authored/teacher-reviewed content; defer AI generation.
OI-3	Parent login in version 1?	Deferred for MVP.
OI-4	Initial subject and concept set	Choose one class, one subject, and a small number of concepts.
OI-5	Initial Intelligence types	Visual, Logical-Mathematical, Linguistic only for MVP
OI-5	Final stack and hosting	React.js, Node.js + Express, Postgres, Render + Cloudinary deployment
OI-6	Exact scoring weights and near-tie margin	Test with sample responses before implementation.
OI-7	Mastery threshold and adaptation rule	Set as configurable pilot parameters; do not claim universal validity.
OI-8	School-visit findings	Add anonymized, relevant observations and requirements.
OI-9	Consent, data retention and deletion process	Confirm with institution before collecting real student data.

Final scope statement

MindBridge is a responsive web application for middle-school students that creates a learning preference profile across eight pathways—Linguistic, Logical-Mathematical, Spatial/Visual, Musical, Bodily-Kinesthetic, Naturalistic, Inter-personal and Intra-personal. It presents approved curriculum concepts through recommended representations, evaluates learning using concept-level baseline and post-assessments, tracks engagement and performance separately, and can recommend a secondary or blended pathway when evidence indicates that adaptation may help. The system supports learning decisions; it does not diagnose or permanently classify student intelligence.